CANONICAL BIJECTION OF SECTIONS SUPPORTED IN A CLOSED SUBSCHEME UNDER PULLBACK

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-etale.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Etale sheaf property on X_etale: sections are determined by their restrictions to the members of an etale covering and glue uniquely from compatible data. invoked at proof lines 278-289 for both the open covering and the etale covering. Provides the two displayed bijections of sections with prescribed support.

Lemma 0.2. *Let X be a scheme. Let $\mathcal{F}$ be an abelian sheaf on $X_{étale}$. Let $\varphi : U' \to U$ be a morphism of $X_{étale}$. Let $Z' \subset U'$ be a closed subscheme such that $Z' \to U' \to U$ is a closed immersion with image $Z \subset U$. Then there is a canonical bijection*

$$\{s \in \mathcal{F}(U) \mid Supp(s) \subset Z\} = \{s' \in \mathcal{F}(U') \mid Supp(s') \subset Z'\}$$

which is given by restriction if $\varphi^{-1}(Z) = Z'$.

Proof. Consider the closed subscheme $Z'' = \varphi^{-1}(Z)$ of U'. Then $Z' \subset Z''$ is closed because Z' is closed in U'. On the other hand, $Z' \to Z''$ is an étale morphism (as a morphism between schemes étale over Z) and hence open. Thus $Z'' = Z' \amalg T$ for some closed subset T. The open covering $U' = (U' \setminus T) \cup (U' \setminus Z')$ shows that

$$\{s' \in \mathcal{F}(U') \mid \mathrm{Supp}(s') \subset Z'\} = \{s' \in \mathcal{F}(U' \setminus T) \mid \mathrm{Supp}(s') \subset Z'\}$$

and the étale covering $\{U' \setminus T \to U, U \setminus Z \to U\}$ shows that

$$\{s \in \mathcal{F}(U) \mid \mathrm{Supp}(s) \subset Z\} = \{s' \in \mathcal{F}(U' \setminus T) \mid \mathrm{Supp}(s') \subset Z'\}$$

This finishes the proof. □

Date: stacks-project:more-etale.tex:local-snapshot-20261012; GNU FDL 1.2 or later.